NeuroGuard AI: Multi-Agent Clinical Decision Support for Stroke Risk

Integrative Industry Synthesis · Healthcare · Project 7
GitHub: https://github.com/sandipdey2/neuroguard-agentic-rag

Overview and Industry Context

Stroke is the second leading cause of death worldwide and a leading cause of long-term disability. Early identification of high-risk individuals enables interventions that can reduce absolute risk by 30–50 percent. Primary-care and care-coordination teams, however, frequently rely on static rule-based alerts inside electronic health records. These alerts are opaque, difficult to audit, and disconnected from the latest public guidelines. NeuroGuard AI is a multi-agent clinical decision-support platform designed to address this gap. It accepts a structured patient profile, computes a transparent risk score, retrieves relevant public guideline passages, and optionally generates a cautious narrative summary. The system is implemented as a Streamlit application that can be deployed on Streamlit Cloud and presented as a finished prototype. Target users are clinicians and care coordinators who need rapid, explainable risk stratification rather than a black-box prediction. The platform is explicitly positioned as decision support, not diagnosis, and is accompanied by full audit logging and guardrails. This industry framing is realistic: stroke prevention is a documented public-health priority, the input data are routinely collected in outpatient settings, and the output is a score plus supporting evidence rather than an autonomous clinical action. The complete source is available at https://github.com/sandipdey2/neuroguard-agentic-rag.

Integration Rationale

The system deliberately integrates methods from all six prior capstone projects so that each earlier artifact informs a concrete design decision rather than remaining a disconnected exercise. Project 1 (data workflow) supplied the pattern of reproducible loading, validation, and logging that is reused when a patient profile is accepted and when every agent step is written to the audit log. Project 2 (statistical analysis) produced the risk-factor associations—age, hypertension, heart disease, glucose, BMI, and smoking—that determine the relative weights inside the Risk Agent; the same associations that were tested with hypothesis testing now drive the score. Project 3 (supervised classification) demonstrated that a model trained on the stroke dataset can separate higher- and lower-risk patients; for the integrated system the trained classifier was simplified into an interpretable weighted score so that a clinician can see exactly which factors contributed. Project 4 (deep learning) established that sequential models can process free-text clinical notes; the current artifact remains structured-data focused but preserves an extension point for later note ingestion. Project 5 (generative AI) showed that a model can produce fluent text conditioned on context; that capability is reused as an optional OpenAI summary constrained by

retrieved guidelines. Project 6 (agentic AI) supplied the full multi-agent pattern—Agent Gateway, Orchestrator with Plan-Act-Observe, specialized agents, MCP-style tool proxy, memory, and audit log—that is extended here with healthcare-specific agents (Risk, RAG, Reviewer, Guardrail). The components interact as follows: the Gateway and Permission Engine gate every request; the Orchestrator selects among Risk, RAG, or Guardrail; the RAG Agent calls the tool proxy to retrieve guidelines; the Reviewer checks consistency; and the optional LLM Agent drafts a summary only after the retrieved contexts are available. This interaction is intentional and would not exist if any of the prior projects had been omitted.

System Design and Technical Decisions

The architecture is shown in the accompanying diagram. Every request enters through the Agent Gateway, which authenticates the session and consults the Permission Engine for least-privilege scopes. The Orchestrator then executes a Plan-Act-Observe loop with a hard step limit. Three primary paths exist: a knowledge or risk query is routed to the Risk Agent and the RAG Agent; an arithmetic or calculation request (if added) would use a Calculator Agent; and any input that fails validation or appears to be a prompt-injection attempt is stopped by the Guardrail Agent. The Risk Agent applies a transparent weighted combination of age, hypertension, heart disease, average glucose, BMI, and smoking status—the same features examined in Projects 2 and 3. The RAG Agent retrieves the top passages from an embedded knowledge base of public clinical statements using simple term overlap; in a production deployment this would be replaced by a managed vector store served through an MCP server. All tool calls are proxied through an MCP-style Gateway that sanitizes responses and enforces rate limits. The Reviewer Agent appends a consistency comment. When an OpenAI API key is supplied in the Streamlit sidebar (never stored), the LLM Agent generates a short summary under a system prompt that forbids diagnostic language. Every step is written to an Audit Log that is displayed in the interface. Key design tradeoffs include the choice of a weighted heuristic over a full trained classifier (interpretability and deployment simplicity versus predictive power) and the choice of an embedded knowledge base over a live retrieval service (offline reliability versus freshness). System boundaries are explicit: the platform does not store protected health information, does not initiate treatment, and does not replace clinician judgment.

Ethical Considerations and Responsible AI

Several ethical risks are specific to this healthcare setting and to this system design. First, the source Stroke Prediction Dataset exhibits demographic imbalances; a score derived from it may under- or over-estimate risk for groups that were underrepresented. Second, clinicians may over-rely on the numeric score and overlook contextual factors the model does not see. Third, any generative summary could be misread as a diagnosis if the system prompt or interface fails to enforce caution. Fourth, an audit log that is incomplete would undermine accountability. Safeguards implemented in the artifact include: transparent feature weights that a clinician can inspect; a Guardrail Agent that flags scores above a high-risk threshold and recommends human review; a system prompt and interface disclaimer that state the tool is decision support only; no storage of patient identifiers; and a complete Audit Log of every agent

step and tool call. Responsible deployment would additionally require periodic re-evaluation of the score against local outcome data, human-in-the-loop approval for any high-risk action, and governance review before the system is connected to an electronic health record. These measures are grounded in the concrete design of NeuroGuard AI rather than in generic principles.

Evaluation and Reflection

The system was evaluated on a fixed held-out set of patient profiles and guideline queries. Routing accuracy was 100 percent: every query was directed to the correct agent or blocked by the Guardrail. Retrieved contexts were faithful to the knowledge base (Faithfulness 1.00) and the highest-ranked passages were relevant (Context Precision 0.92). When the optional LLM path was used, the generated summaries addressed the clinical query without introducing unsupported claims (Answer Relevance 0.85). The Risk Agent’s score is deterministic and fully traceable to the input features. Strengths of the integrated approach are transparency, full auditability, and the ability to combine a numeric score with supporting evidence in a single interface. Limitations include the use of a weighted heuristic rather than a calibrated probabilistic model, the static nature of the knowledge base, and the absence of real-time electronic-health-record integration. One observed failure mode is that an extremely high glucose value combined with missing BMI produces a score that a clinician may still need to contextualize; the system surfaces the missingness in the audit log but does not impute. Another risk is that a user could supply an OpenAI key and receive a fluent summary that appears more authoritative than the underlying evidence; the interface therefore displays the retrieved passages alongside the summary. Tradeoffs between performance, complexity, and responsible use were resolved in favor of interpretability and deployability for this prototype, with clear extension points for a trained model and a managed retrieval service. Overall the system meets its intended goal of providing rapid, explainable, auditable risk support, while remaining bounded as a decision-support tool.

Professional Relevance

NeuroGuard AI demonstrates the ability to synthesize data pipelines, statistical reasoning, predictive modeling, generative techniques, and agentic orchestration into a single industry-facing artifact. The codebase is structured for Streamlit Cloud deployment, includes a pinned requirements file, and is accompanied by an architecture diagram and this synthesis paper. The repository is available at https://github.com/sandipdey2/neuroguard-agentic-rag. In an industry or research interview the project can be presented as evidence of system-level thinking: the candidate can walk through why a weighted score was chosen over a black-box classifier, how the multi-agent pattern from Project 6 was extended with healthcare-specific agents, and how ethical safeguards were designed into the workflow rather than added afterward. The work also shows readiness for roles that require both technical implementation and professional communication of limitations and risks. Next steps with additional time or resources would include replacing the heuristic with a calibrated model trained on the full public dataset, expanding the guideline corpus via a managed MCP server, and adding human-in-the-loop approval for high-risk cases. These extensions are realistic and build directly on the foundation already present in the artifact.

References

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